# Laboratory Report

*Digital Controller Implementation*

EEE4118F: Process Control & Instrumentation

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# Contents

**Abstract**

In this report, the design and implementation of a digital control system for a DC servo motor is presented. The aim of this laboratory was to achieve precise and robust position setpoint tracking along with control of the motor's speed response under varying plant loading and braking conditions. This was accomplished by means of cascaded digital control, with a proportional controller of $K_v = 2$ in the inner (velocity) loop and a lead compensator with a transfer function given by $C_p(z) = 0.5 \times \frac{13.36z-10.67}{z-0.6577}$ in the outer loop. The system displayed zero steady-state error and an overshooot of $M_p \leq 20\%$ in tracking position setpoints within its operating range. Furthermore, the system response settled to within 2% of the reference position within 2 seconds and exhibited output disturbance rejection – returning to the reference position within 3 seconds.

# 1 Introduction

Due to advances made in microcomputers, digital controllers have become very popular [1]. The use of digital computers in control system yields several advantages over analog systems, such as: reduced cost, flexibility in response to design changes and noise immunity [2]. But we live in an analog world; physical systems and processes occur in continuous-time and produce continuous values [3]. Hence the DC servo motor digital control system in question contains both analog and digital components, namely the plant (and its auxiliary components) and the digital controller, respectively. Therefore, means for conversion from analog to digital (D/A) and digital to analog (A/D) are required for each subsystem to interface with another [2] as shown in Fig.1 below.

A continuous-time signal is represented by a sequence constisting of samples of the signal equally spaced in time by a sampling period $T_s$. This "sampled-time" sequence is represented by its $z$-transform in the digital domain [4] where the computer can work with it to implement the digital control algorithm.

While the digital controller can be designed in the $z$-domain [5], bilinear transforms give us the ability to apply s-plane design techniques to digital systems [2]. In this laboratory, the digital controller was designed using a bilinear transform, namely the Tustin transform [6] of the $z$-domain representation where $w = \frac{2}{T_s}\frac{z-1}{z+1}$. The design of the digital controller in this intermidiate $w$-domain is beneficial because the $w$-domain and the $s$-domain behaviour of a continuous-time plant are comparable in the low frequencies hence $s$-domain specifications can be used to derive $w$-domain specifications [7].

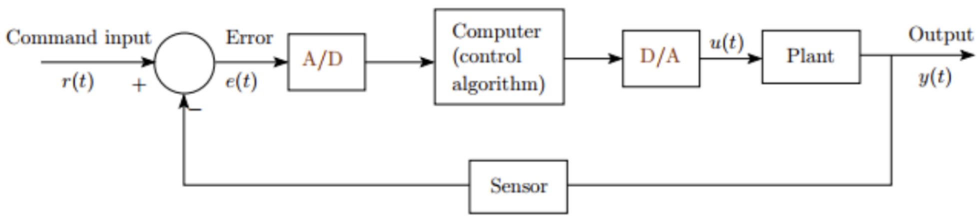


Figure 1: Block diagram of a general digital control system for an analog plant. Adapted from Digital Control Systems lecture notes, courtesy of GVP College of Engineering.

## 2 Design Specifications

The DC servo motor control system was designed to satisfy the following performance specifications[1]:

| ID | Specification | Value | Description |
|---|---|---|---|
| SS1 | Position Setpoint Tracking | $e_{ss} \approx 0\%$ | Zero steady-state error trackingof position setpoints. |
| ST1 | Settling Time | $t_{2\%} \leq 2$ s | Position output settles within 2% of its final value within 1 second. |
| OS1 | Damping Specification | $M_p \leq 20\%$ | A peak overshoot of less than 20% above the reference value. |
| DR1 | Disturbance Rejection | $t_d \leq 2$ s | The system response returns to the reference value within 1 second. |

Table 1: Performance specifications for the DC servo motor control system.

Furthermore it was specified that the $s$-domain design must have a minimum phase margin of $\phi_d = 15°$ to maintain stability after discretisation.

## 3 System Modelling

The operation of a DC servo motor is governed by a set of coupled differential equations. Equation (1) describes the electrical characteristics of the motor and is derived from Kirchhoff's voltage law applied to the armature circuit shown in Figure 2.

$$v_a(t) = R_a\, i_a(t) + L_a \frac{d}{dt} i_a(t) + K_b\, \omega_m(t) \tag{1}$$

Equation (2) describes the mechanical characteristics of the motor and is derived from Newton's second law applied to the rotational motion of the motor shaft.

$$J \frac{d}{dt} \omega_m(t) + \beta\, \omega_m(t) = K_t\, i_a(t) \tag{2}$$

Where:

| Symbol | Name | Units |
|---|---|---|
| $i_a$ | Armature current | A |
| $R_a$ | Armature winding resistance | Ω |
| $L_a$ | Armature winding inductance | H |
| $K_b$ | Back-EMF constant | V·s/rad |
| $K_t$ | Torque constant | N·m/A |
| $J$ | Equivalent inertia | kg·m$^2$ |
| $\beta$ | Viscous damping | N·m·s/rad |

Table 2: Servo-motor Parameters.

[1] An error within $\varepsilon < 0.1$ is tolerated and passed as perfect tracking.

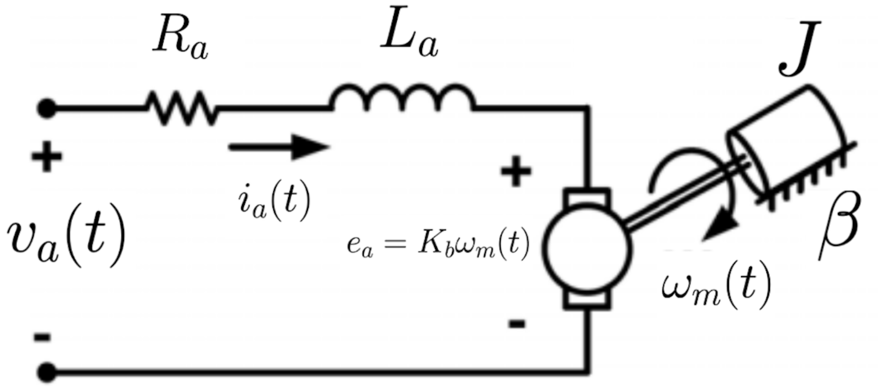


Figure 2: Equivalent circuit of a DC servo motor [8].

From this, the transfer function relating the motor's angular velocity $\omega_m(s)$ to the applied armature voltage $V_a(s)$ was found to be:

$$P(s) = \frac{\omega_m(s)}{V_a(s)} = \frac{K_\phi/R_a}{Js + \left(\frac{K_\phi^2}{R_a} + \beta\right)} \tag{3}$$

The plant speed model $P(s)$ takes the form of a first-order system:

$$P(s) = \frac{A}{\tau s + 1}\,, \quad \text{where} \quad A = \frac{K_\phi/R_a}{(K_\phi^2/R_a) + \beta} \quad \text{and} \quad \tau = \frac{J}{(K_\phi^2/R_a) + \beta} \tag{4}$$

The motor was allowed to run under varying conditons of loading and braking. This was achieved by adding or removing metal discs to the motor shaft and by applying an eddy current brake, respectively. Consequently, the system parameters $J$ and $\beta$ varied, inducing an apparent uncertainty in the effective plant steady-state gain $A$ and time constant $\tau$.

# 4 System Identification

Here, the process of obtaining the parameters of the DC servo motor first-order speed model by least-squares is presented.

## 4.1 Least-sqaures Solution

It is known that the plant speed model takes the form:

$$\frac{\omega_m(s)}{V_a(s)} = \frac{A}{\tau s + 1} \tag{5}$$

Multiplying both sides by $(\tau s + 1)$ and rearranging gives:

$$\omega_m(s) = A \cdot V_a(s) - \tau s \cdot \omega_m(s) \tag{6}$$

Applying the inverse Laplace transform ($\mathcal{L}^{-1}$) to both sides gives:

$$\omega_m(t) = A \cdot v_a(t) - \tau \cdot \frac{d}{dt}\omega_m(t) \tag{7}$$

It is important to note that $\omega_m(t)$ and $v_a(t)$ are measured signals. The least-sqares solution may be obtained from the above differential equation, but the differentiation of measured signals tends to amplify high frequency noise [9]. Hence, the least-squares solution is obtained from re-writing equation (7) in integral form as follows:

$$\int_0^t \omega_m(\lambda)d\lambda = A \cdot \int_0^t v_a(\lambda)d\lambda - \tau \cdot \omega_m(t) + \tau \cdot \omega_m(0) \tag{8}$$

Equation (8) above may be written in matrix form as:

$$\begin{bmatrix} \int_0^t \omega_m(\lambda)d\lambda \end{bmatrix} = \begin{bmatrix} \int_0^t v_a(\lambda)d\lambda & \omega_m(t) & 1 \end{bmatrix} \begin{bmatrix} A \\ -\tau \\ \tau \cdot \omega_m(0) \end{bmatrix} \tag{9}$$

Equation (9) may be written in the form $\mathbf{y} = \mathbf{M}\boldsymbol{x}$, a linear regression problem. An error function is defined as

$$\boldsymbol{\epsilon} = \mathbf{y} - \mathbf{M}\boldsymbol{x} \tag{10}$$

where $\mathbf{y}$ is obtained from the integration of the measured $\omega_m(t)$ signal, $\mathbf{M}$ is obtained from the integration of the measured $v_a(t)$ signal and the measured $\omega_m(t)$ signal, and $\boldsymbol{x}$ is the vector of unknown parameters. The square of the error function is:

$$\boldsymbol{\epsilon}^T\boldsymbol{\epsilon} = \boldsymbol{x}^T\mathbf{M}^T\mathbf{M}\boldsymbol{x} - 2\boldsymbol{x}^T\mathbf{M}^T\mathbf{y} + \mathbf{y}^T\mathbf{y} \tag{11}$$

and we find the minimum by differentiating with respect to $\boldsymbol{x}$ and setting the result to zero for a turning point:

$$\frac{\partial}{\partial \boldsymbol{x}}\left(\boldsymbol{\epsilon}^T\boldsymbol{\epsilon}\right) = 2\mathbf{M}^T\mathbf{M}\boldsymbol{x} - 2\mathbf{M}^T\mathbf{y} = 0 \tag{12}$$

which gives the least-squares solution as:

$$\boldsymbol{x} = \left(\mathbf{M}^T\mathbf{M}\right)^{-1}\mathbf{M}^T\mathbf{y} \tag{13}$$

## 4.2 Results

The results are presented in the table below. The measured step resonse[2] along with the model repsonse for task 1 are shown, plotted on the same graph in Fig.3.

| Task No. | **Motor Load** | **Braking** | **Steady-state Gain ($A$)** | **Time Constant $\tau$ (s)** |
|---|---|---|---|---|
| 1 | Light | None | 13.4 | 0.563 |
| 2 | Light | Light | 6.84 | 0.217 |
| 3 | Light | Heavy | 5.43 | 0.178 |
| 4 | Heavy | None | 15.2 | 1.920 |

Table 3: Parameters of the DC servo motor speed model obtained by least-squares system identification.

[2] In response to a step signal of amplitude 0.6V.

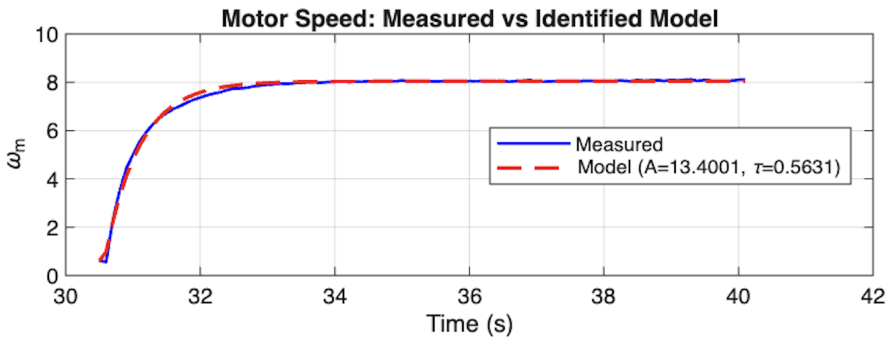


Figure 3: Measured step response and model response for task 1.

From the results in Table 3, the identified first-order speed model of the DC servo motor was found to be:

$$P(s) = \frac{A}{\tau s + 1} \text{ , where } A \in [5.43, 15.2] \quad \text{and} \quad \tau \in [0.178, 1.920] \text{ s} \tag{14}$$

### 4.3 Discussion

The results of the system identification process are consistent with the expected behaviour of the system from equation (4). Comparing tasks 2 and 3 (where the eddy current brake was applied) to tasks 1 and 4 shows a decrease in the steady-state gain, corresponding to increased damping ($\beta$). Comparing tasks 1 and 4 where the damping is constant but the load is increased, shows an expected increase in the time constant $\tau$ due to the increase in inertia ($J$). The system modelling process has clearly indicated how the changes in motor parameters $J$ and $\beta$ impact the behaviour of a DC servo motor.

## 5 Controller Design

The closed-loop simulink model of the system used for controller design is shown in the figure below:

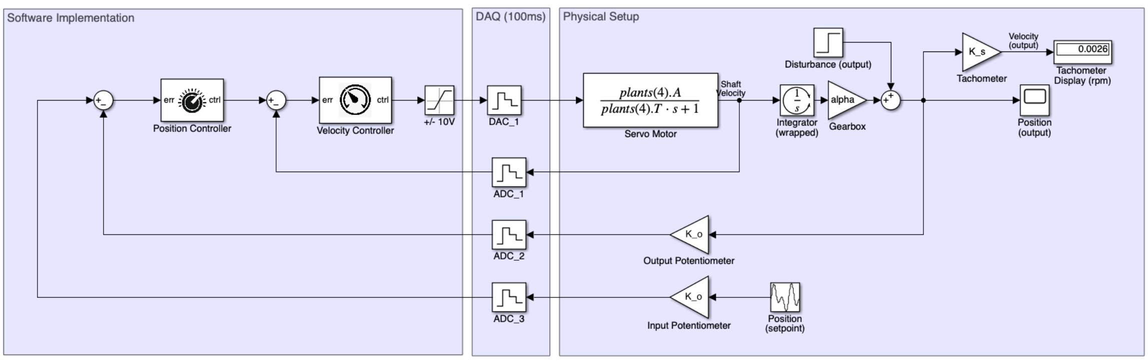


Figure 4: Simulink Block Diagram of the Closed-loop Cascaded Control System

This diagram shows how the physical system interfaces with software for the implementation of control alogrithms. The controller is designed in the intermediate $w$-domain as outlined in Section 1.

## 5.1 Continuous-time Plant in the w-domain

The exact $w$-domain plant is obtained by running the MATLAB command below:

```
% Direct transform using the double-conversion method
P_w = d2c(c2d(P_s, T, 'zoh'), 'tustin');
```

Listing 1: Bilinear (Tustin) Transform via ZOH discretization

## 5.2 Design Procedure

The design procedure used in this lab is that which is advocated for by E. Boje in [7]. The procedure is oulined and followed below:

### 5.2.1 Obtain w-domain specifications

Because the sampling rate is much greater than the highest plant corner frequency, the $w$-domain specifications are considered equal to the $s$-domain specifications [7]. The $s$-domain specifications are derived from the performance sepcifications presented in Table 1:

#### Velocity Controller/Bandwidth Specification

It is apparent (with reference to Table 1) that the performance of the system is primarily concerned with robust postion tracking as opposed to speed control. This gives us a reason not to overdesign the velocity controller and to focus time, energy and most importantly, resources on what's most important – giving the client what they have asked for.

In order to have an effective cascade control system, it is essential that the inner loop responds much faster than the outer loop [10]. The secondary process (velocity control) must react to the secondary controller's efforts at least three or four times faster than the primary process (postion control) reacts to the primary controller. This allows the secondary controller enough time to compensate for inner loop disturbances before they can affect the primary process [11]. The inner loop dynamics then appear as a unity gain to the outer loop, simplifying the design of the primary controller. The velocity controller is selected[3] to be

$$K_v = 2 \tag{15}$$

and the inner loop frequency response for all plant loading and braking conditions is shown in the figure below. The worst-case bandwidth occurs in Task 4 as shown in Figure 5 and is found to be $\omega_{\mathrm{BW(inner)}} = 15.2$ rad/s. For this project we would like the inner loop bandwidth to be at least 5 times larger than the outer loop bandwidth, that is:

$$\begin{aligned} \omega_{\mathrm{BW(inner)}} &\geq 5 \times \omega_{\mathrm{BW(outer)}} \\ \omega_{\mathrm{BW(outer)}} &\leq 3.04 \text{ rad/s} \end{aligned} \tag{16}$$

Therefore the position loop crossover frequency should be no greater than 3.04 rad/s [2].

[3] The value of $K_v$ was selected through an iterated design process as a result of limited bandwidth problems encountered further along in the design.

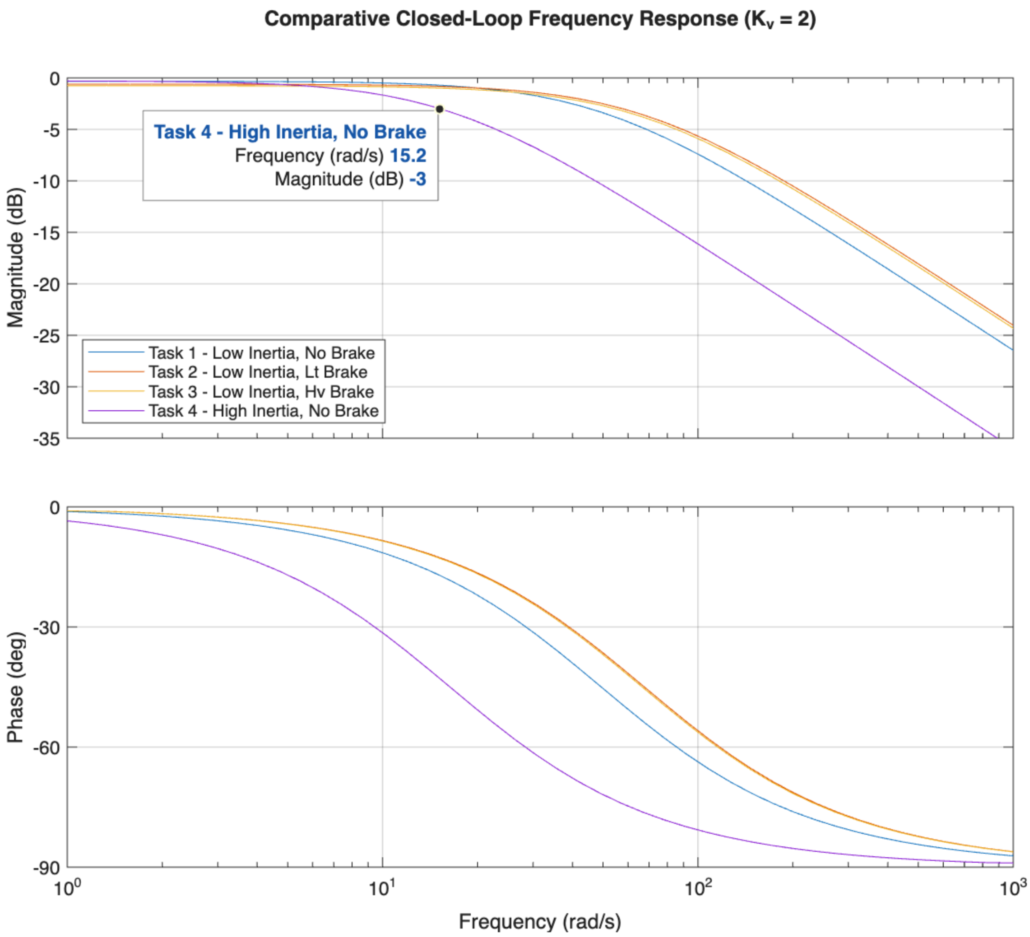


Figure 5: Bode Plots for the Inner Loop Frequency Response Across All Plant Loading and Braking Conditions

### Steady-state Error (SS1)

As shown in Figure 4, the position model of the plant is obtained by appending an integrator term onto the speed model. The second-order position model of the plant is given by:

$$P_\phi(s) = \frac{\alpha}{s} \cdot P_v(s) \tag{17}$$

where $\alpha$ is an integration constant determined experimentally to be $\alpha = 8.11$ rad/rad. The loop transfer function for the position control system without a postion controller is:

$$L_\phi(s) = \frac{\alpha}{s} \cdot \frac{K_v \cdot A}{\tau s + 1} \tag{18}$$

The steady-state error is obtained by applying the Final-Value Theorem [2] on the closed-loop response to a step input of amplitude $\gamma$:

$$\begin{aligned} y(\infty) &= \lim_{s \to 0} s \cdot \frac{\gamma}{s} \cdot \frac{L_\phi(s)}{1 + L_\phi(s)} \\ &= \gamma \cdot \lim_{s \to 0} \frac{L_\phi(s)}{1 + L_\phi(s)} \\ &= \gamma \cdot 1 = \gamma \end{aligned} \tag{19}$$

Due to the integrator in the position model, the system theoretically tracks postion setpoints perfectly in steady-state i.e $e_{ss} = 0$.

### Settling Time (ST1)

The 2% settling time of the system ($t_{2\%}$) is related to the bandwidth and the damping coefficient by [2]:

$$\omega_{\text{BW}} = \frac{4}{t_{2\%}\zeta}\sqrt{(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}} \tag{20}$$

Given a maximum bandwidth of $\omega_{\text{BW}} = 3.04$ rad/s in equation (16) and a maximum 2% settling time of 2 seconds, the minimum[4] damping coefficient was found to be:

$$\zeta_{\text{min}} = 0.682 \tag{21}$$

The phase margin $\phi_M$ at crossover frequency is related to the damping coefficient by:

$$\phi_M = \tan^{-1}\left(\frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+4\zeta^4}}}\right) \tag{22}$$

Phase margin increases with damping [2], hence the worst-case phase margin occurs at the lowest tolerable damping $\zeta_{\text{min}} = 0.682$. Substituting this value into equation (22) yields a minimum phase margin of:

$$\phi_M = 64.2° \tag{23}$$

The relationship between the phase margin and the peak closed-loop magnitude $M_p$ is approximately:

$$M_p \approx \frac{1}{2\sin(\phi_M/2)} \tag{24}$$

hence, placing a lower limit of $\phi_M \geq 64.2°$ implies:

$$M_p \lesssim 0.94 \tag{25}$$

Since $M_p$ is the peak magnitude of the closed loop system, equation (25) can be translated as

$$\left|\frac{L}{1+L}\right| \leq -0.53 \text{ dB } ,\omega \leq 3.04 \text{ rad/s} \tag{26}$$

### Overshoot (OS1)

For a percentage overshoot $M_p \leq 20\%$, the corresponding damping coefficient[5] specification is $\zeta \geq 0.456$. This is less than the damping required to satisfy the settling time and bandwidth constraints simultaneously hence this specification is neglected in favour of the settling time specification which ensures a overshoot well below 20%.

[4] Equation (20) was plotted in Desmos to find that maximising bandwidth and settling time as per our constraints, minimises damping.

[5] using the formula $\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2+\ln^2(\%OS/100)}}$ from [2].

**Disturbance Rejection**

The transfer function of output disturbance to the plant output with feedback is

$$T_{y/d} = \frac{1}{1+L} \tag{27}$$

This yields a second order high-pass of the form

$$\frac{1}{1+L} = \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{28}$$

with a 2% percent settling time approximated by

$$t_{2\%} \approx \frac{4}{\omega_{\mathrm{BW(cl)}}} \tag{29}$$

where $\omega_{\mathrm{BW(cl)}}$ is the closed-loop bandwidth of the loop. Substituting $t_{2\%} = 2$ s into the above equation yields $\omega_{\mathrm{BW(cl)}} \approx 2$ rad/s, which is acceptable for the design of the outer loop (ref: eq (16)). Disturbance rejection is achieved within the control bandwidth [12] and to achieve this, from equation (28), we can make use of a sensitivity goal [12]. The maximum tolerable error in this lab is 0.1 (steady-state error) which corresponds to -20 dB attenuation of external disturbances. Hence to acheive the required disturbance rejection sepcification, within a fraction of the derived bandwidth, the following constraint is imposed on the system design:

$$\left|\frac{1}{1+L}\right| \leq -20 \text{ dB} \;, \omega \leq 0.5 \text{ rad/s} \tag{30}$$

### 5.2.2 QFT Controller Design

**Plant Template**

A QFT design approach is taken to design a controller for the uncertain plant model presented in equation (14). The first step is to generate plant's frequency response set i.e templates [13]. The frequency response set of the plant is presented in the figures below[6]:

With reference to Figure 7, at higher frequencies it is apparent that one plant model may be appropriately selected to be the nominal plant due to being much closer to the stability boundary than others. This nominal plant is identified to be the loading/braking condition of Task 4:

$$P_0(w) \approx \frac{15.2}{1.920w + 1}\left(1 - \frac{wT}{2}\right) \tag{31}$$

[6]The QFT design is done using a a MATLAB plugin developed by Terasoft, Inc. Visit `https://www.terasoft.com/products/QFT/index.html` for more information.

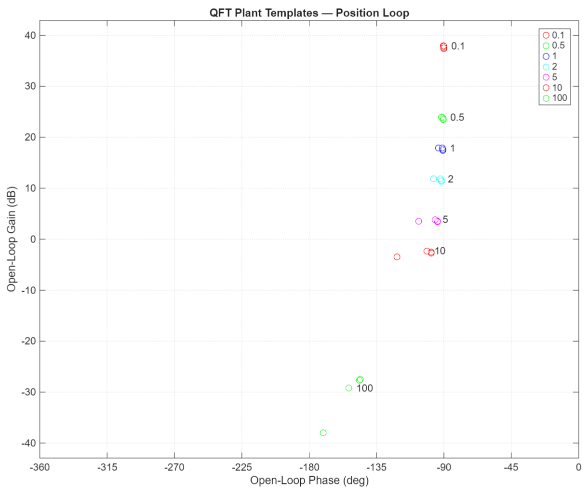


Figure 6: Plant Templates Showing the Frequency Response Set from 0.1 to 100 rad/s

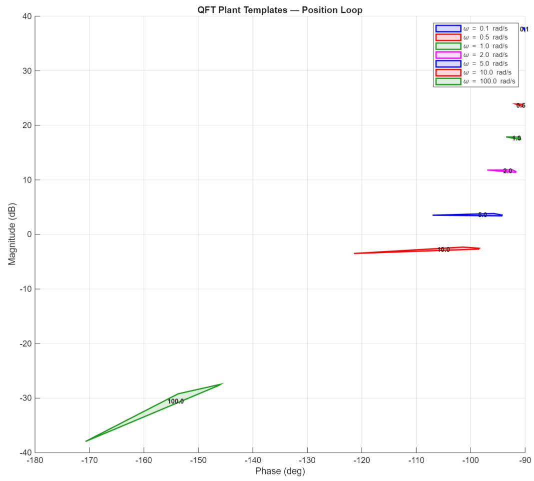


Figure 7: Shaded templates for increased visibility

### Nominal Bounds

The next step is to generate QFT bounds from the design specifications. With reference to equations (26) and (30), the bounds are implemented in MATLAB using the code snippet below:

```
% Settling time bound
w = [0.1, 0.5, 1, 2, 5, 10]; % Select frequencies
bnd_st = sisobnds(1, w , 0.94, p);
% Disturbance rejection bound
bnd_dr = sisobnds(2, w, 0.1, p);
% Robust Stability
bnd_rs = sisobnds(2, w, 3, p);
```

Listing 2: Implementing the QFT Design Bounds using the QFT Toolbox in MATLAB

The addition of lead/lag compensators to alter the frequency response can potentially destabilise the system. For this reason, a robust stability specification is imposed onto the controller design to ensure safety and stability across all frequencies:

### QFT Loop Shaping

The QFT loopshaping is done and illustrated in the figures 8 and 9 below. The controller is extracted directly from the design tool. As expected, a lead compensator (and proportional gain) was required to meet the QFT design specifications. The position controller was found to be:

$$C_p(w) = 14.5 \cdot \frac{w + 2.238}{w + 4.13} \tag{32}$$

This added 17.3° of additional phase at 3.04 rad/s.

### 5.2.3 Digital Control Algorithm

The final step is to turn the $w$-domain controller into an algorithm for digital control via the inverse Tustin trasnform [7].

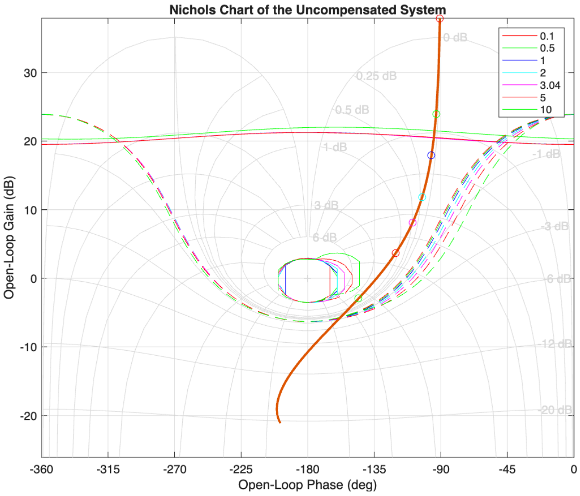


Figure 8: Nichols Plot Showing the Uncompensated System and the QFT bounds

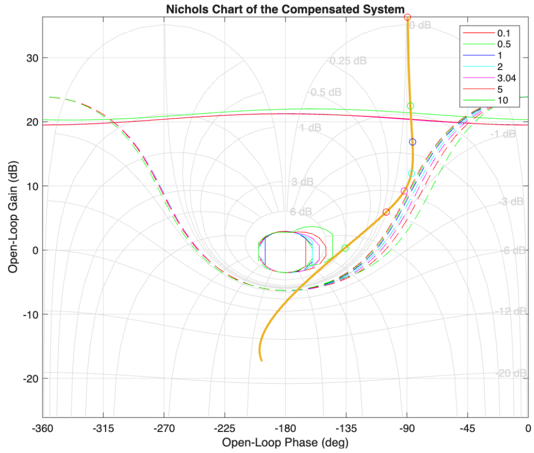


Figure 9: Nichols Plot Showing the Compensated System and the QFT bounds

The $z$-domain position controller is found to be:

$$C_p(z) = \frac{13.36z - 10.67}{z - 0.6577} \tag{33}$$

The controllers are implemented and tested in Simulink for system validation.

# 6 System Validation

The complete cascaded control system for the DC servo motor drive is shown in the Simulink block diagram in Figure 4. The system is subject to a 1V step input at the position setpoint 1 second and a step output disturbance of 1V is applied at 5 seconds. The response is observed for 10 seconds and plotted in Figure 10.

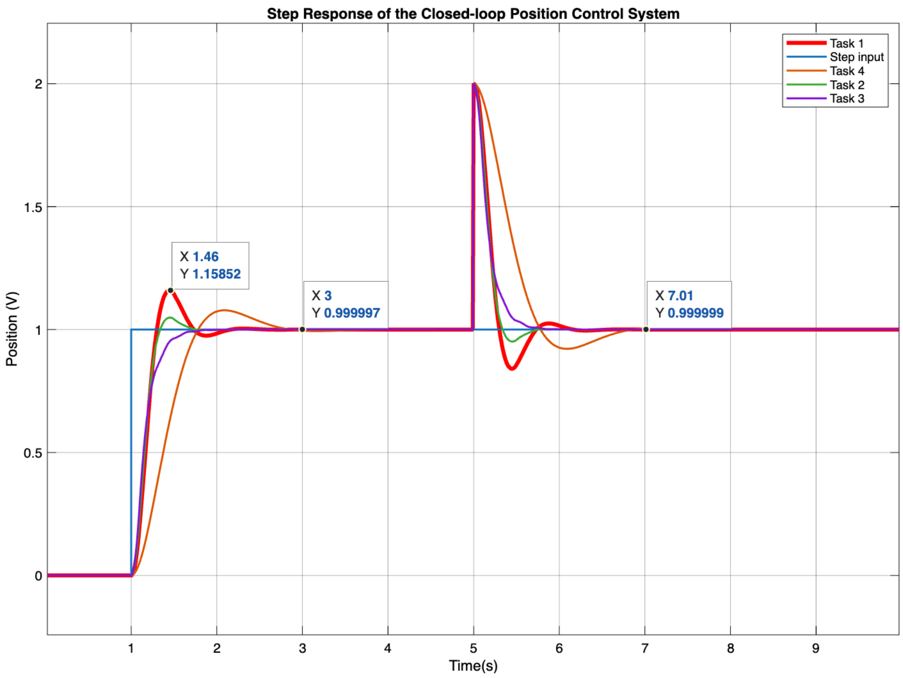


Figure 10: Validation Results of the Cascaded Control System

It is important to note that some fine-tuning was done on the position controller to achieve the performance shown in the figure above. A gain of 0.5 was applied to the position error before it is fed into the position controller. This reuslted in a reduction in peak overshoot from 40% to 16% in the worst case (Task 1).

The model shows zero steady-state error across all plant loading and braking conditions. The slowest system (Task 4) exhibits a 2% settling time below 2 seconds, and likewise all responses return to the setpoint within 2 seconds from the application of a step disturbance. The highest peak overshoot is approximately 16%, corresponding to Task 1.

# 7 Controller Implementation

## 7.1 Software Implementation

Discrete-time systems are implemented using constant coefficient difference equations [14] hence the controller is written in terms of delayed input and ouput terms.

$$\frac{U(z)}{E(z)} = \frac{13.36 - 10.67z^{-1}}{1 - 0.6577z^{-1}} \tag{34}$$

The input and output terms are separated:

$$U(z)(1 - 0.6577z^{-1}) = E(z)(13.36 - 10.67z^{-1}) \tag{35}$$

The inverse $z$-transform is applied:

$$u[k] - 0.6577u[k-1] = 13.36e[k] - 10.67e[k-1] \tag{36}$$

Then expressing the output $u[k]$ in terms of the input and past outputs:

$$u[k] = 0.6577u[k-1] + 13.36e[k] - 10.67e[k-1] \tag{37}$$

The control algorithm is implemented in C++ and the code snippet is shown in the listing below:

```
double offError = PosError - 0.25; // position error with fixed offset
// Position Controller C_p(z) ======================
PosOutput = (0.6577 * PrevPosOutput) + (13.36 * offError) + (-10.67 *
    PrevPosError);
//  Velocity Controller K_v = 2 ====================
double VelError = 0.5 * PosOutput - Velocity; // fine-tuning
ut_Output = 2.0 * VelError;
PrevPosOutput = PosOutput;
PrevPosError  = offError;
```

Listing 3: Discrete Implementation of the Cascaded Position and Velocity Controllers

## 7.2 Results

The real-time tracking performance of the control system was evaluated on the physical DC servo motor setup. The resulting plot is shown in Figure 11.

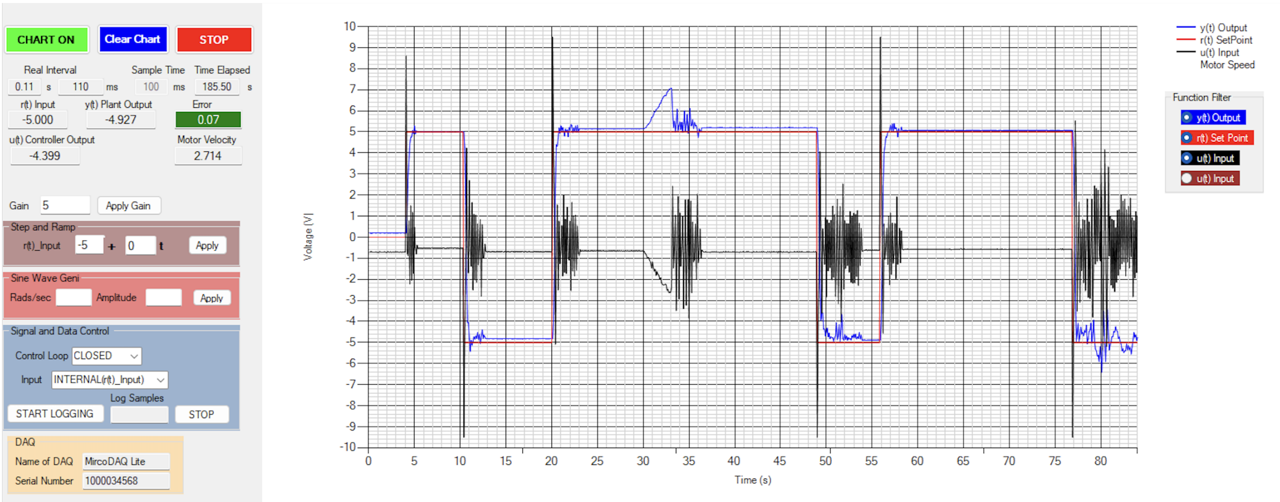


Figure 11: Real-Time Tracking Performance of the Cascaded Control System

The position error was consistently within the tolerable range of $\varepsilon < 0.1$ across all tasks, confirming that the system achieved zero steady-state error tracking of position setpoints. The system response settled within 2% of the reference position within 2 seconds and exhibited a peak overshoot of less than 20%, satisfying the design specifications outlined in Section 2. Furthermore, the system demonstrated effective disturbance rejection, returning to the reference position within 3 seconds after a step disturbance was applied, failing to meet the 2-second requirement by 1 second but maintaining tracking performance. The robustness of the system was further tested by changing the attenuation factor of the power amplifier in the lab from 4 to 5, which resulted in sustained oscillations as shown in the figure above at $t \approx 76$ s.

# 8 Conclusion & Recommendations

In this report, a cascaded digital control system for a DC servo motor was designed, implemented and validated against strict performance specifications. The system achieved plausible position setpoint tracking and fast transient repsonse. However, the robustness of the system was not completely satisfactory due to a slow disturbance rejection response and sustained oscillations when the power amplifier attenuation factor was changed.

## 8.1 Recommendations

The system model could have been improved by taking into account the input and output potentiometers along with tachometer feedback. Furthermore, changes in the amplifier attenuation factor could have been included during the system identification process to improve overall robustness.

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